



Vortex induced vibrations of a bridge deck: Dynamic response and surface pressure distribution



M. Belloli, F. Fossati, S. Giappino, S. Muggiasca*

Politecnico di Milano, Dipartimento di Meccanica, Milano, Italy

ARTICLE INFO

Keywords:

Vortex shedding
Vortex induced vibration
Long span bridge
Multi-box girder
Wind tunnel test

ABSTRACT

This paper describes selected results collected during the wind tunnel tests carried out at Politecnico di Milano Wind Tunnel on the deck sectional model of the Messina Strait Bridge. It is a long span suspension bridge characterized by a multi box deck. Aerodynamic forces and surface pressure distributions have been measured in order to characterize static and dynamic response of the deck. Vortex induced vibrations phenomenon has been studied with particular attention: it represents one of the most important aspects to be considered while designing the deck shape of long span bridges and it has to be carefully analyzed. In fact, even if it can be separated from the stability problem for the modern bridges, VIV, in case of low Scruton numbers, can lead the deck to reach very large vibration amplitudes, also at very low wind speeds.

© 2014 Elsevier Ltd. All rights reserved.

1. Introduction

Nowadays longer and longer spans have been realized for suspension and cable stayed bridges designing slender and flexible structures. This kind of structures showed an higher sensitivity to the wind excitation. In particular, vibrations induced by vortex shedding phenomena can become relevant at moderate wind velocities as observed in different kind of flexible structure (Belloli et al., 2011, 2012; Fossati et al., 2010; Zasso et al., 2008). In order to avoid higher level of vibration, the aerodynamic design of a bridge deck has to be taken in strong consideration. Indeed, a correct definition of the deck shape is surely preferable than adding devices to control the phenomenon. In Larsen et al. (2000), the effect of guide vanes to mitigate vortex shedding is showed: this solution permits to solve the problem in an already built bridge but it requires more money and maintenance than to correctly projected in advance the deck from an aerodynamic point of view. An example of a correct shaping on a trapezoidal box girder bridge deck is shown in Larsen and Wall (2012). In this paper the effect of the angle between the horizontal bottom plate and the lower inclined panels is investigated: it has been observed that values equal to 26 deg permit vortex to develop so that vortex shedding occurs, whereas for lower angles the phenomenon does not appear. A few tests were also performed on twin-deck bridge: multi-box solution is preferable in terms of aerodynamic stability because shows higher flutter velocity, on the other hand the fluid–structure

interaction become more complex. In particular the mechanism that produce vibration due to vortex shedding results more complicated due to the wake effects of the windward deck on the leeward one. In Kwok et al. (2012) a sectional model of a twin-deck bridge in bare configuration has been observed changing the gap width and observing the effects on vortex shedding. While a single deck bridge experiences only one type of vortex shedding a multi-deck bridge can be subjected to more than one type of vortex shedding mechanisms. Also in Li et al. (2011) and in Laima et al. (2013) the vortex shedding phenomenon on twin-deck bridge is investigated highlighting the main role played by the gap. Moreover, this subject has been discussed also in developing numerical models able to reproduce vortex-shedding induced vibration on long-span bridges in the context of nonlinear finite-element modeling for dynamic analyses (Caracoglia et al. (2008)).

The interest in studying the phenomenon is finalized to control or eliminate the vibrations induced by vortex shedding: avoiding significant level of vibration it is important not only for the structure safety but also for the comfort and the safety sensation of the drivers on the bridge. In these studies a key role is played by the wind tunnel tests, that are finally become a standard tool in bridge design.

This paper describes selected results collected during the wind tunnel tests carried out at Politecnico di Milano Wind Tunnel on deck sectional model of the Messina Strait Bridge. It is a long span suspension bridge with a main span of 3300 m. This bridge is an example of a multi-box bridge with two road box girder and one railway box girder in the middle (see Fig. 1). The three separated decks are connected together every 30 m, along the longitudinal direction with a transversal beam having the function of structural connections. The complex fluid–structure interaction of this bridge

* Corresponding author.

E-mail address: sara.muggiasca@polimi.it (S. Muggiasca).

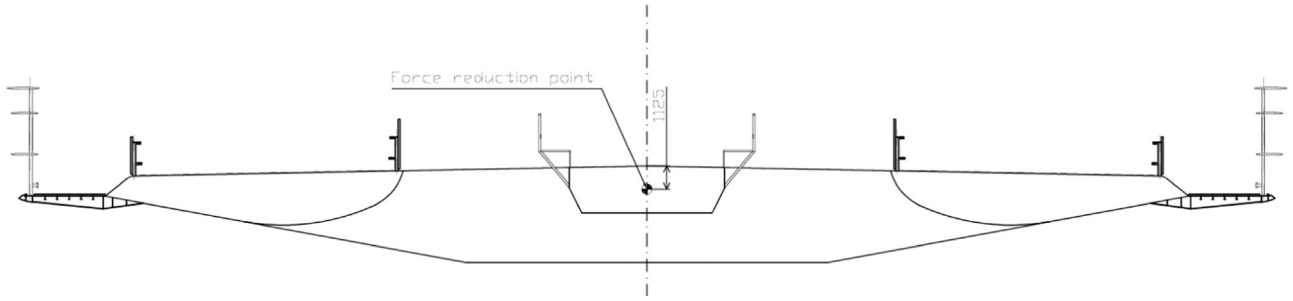


Fig. 1. Messina Strait Bridge deck.

was studied measuring aerodynamic forces and surface pressure distributions: a particular attention was devoted static and dynamic response of the deck when vortex induced vibrations occur. The deck configuration analyzed is characterized by two rounded roadway girders and a central sharp edge rail box. Previous studies (Diana et al., 2004, 2006) highlighted that a trip wire or a sharp edge in the lower part of road boxes fix the flow separation point increasing VIV. Only the railway box shape effects was investigated.

2. Experimental set-up

The experiments were performed at Politecnico di Milano wind tunnel: the aerodynamic characterization of the multi-box deck of Messina Strait Bridge was carried out using two rigid sectional models 1:45 scaled (MB01 and MB02). They were 1.33 m wide (B) and 4 m long (L) resulting in an aspect-ratio of 3. Both the models were carbon fibre manufactured for lightness and stiffness purposes, with fittings (wind screens, barriers etc.) glued on. The models differed only for the shape of the rail box girder, in particular the angle of the lower inclined panels assume different values as shown in Figs. 2 and 3. As observed in Larsen and Wall (2012) the angle of inclined panels can strongly influence the vortex generation in particular an intermediate level of inclination can be considered critical. The models were equipped by pressure measurement devices in order to measure surface pressure distributions in a cross section placed in the middle between two following transverses. The measuring points (60 pressure taps), shown in detail in Fig. 4 (left), are connected to high sample-rate multi-channel pressure scanners placed inside the model transverses (PSI ESP 32 channels). Fig. 4 (right) shows the taps distributions on the downwind roadway girder, a similar distribution was realized on the upwind girder. It is possible to note that a relevant number of pressure taps were placed in the lower side of the roadway box girders. Pressure measurements are very useful to better understand the flow structure interaction as observed in Belloli et al. (2012), Zasso et al. (2008), Argentini et al. (2012).

Pressure distributions were provided in terms of non dimensional pressure coefficients defined as:

$$C_p = \frac{P_i - P_s}{q} \quad (1)$$

where P_i is the actual pressure measured at each tap P_s is reference static pressure and $q = 1/2\rho V^2$ is the dynamic pressure of the incoming wind.

By the integration of the surface pressures over a single girder it was possible to evaluate the forces on each roadway girder separately. It is to note that those data can be considered representative of the aerodynamic behaviour of the deck even if they differ from the whole forces because they do not take into account wind forces on fittings. They are substantially dependent on the pressures distributions on the lower part of each girder that is strongly influenced by vortex shedding phenomena.

The models were tested in two different conditions:

- (1) Deck sectional models mounted on an external force balance to measure the aerodynamic forces as a function of the angle of attack. These models were instrumented with pressure taps to define the pressure distribution on the external surface (see Fig. 5 (left)).
- (2) Deck sectional models elastically suspended to the wind tunnel walls to investigate vortex shedding phenomena. Dynamic responses of the models and pressure distributions were measured (see Fig. 5 (right)).

Set-up 1 was mounted in the high velocity tests section of the wind tunnel: this test section is 4 m wide and 3.84 m high. The maximum wind velocity is 55 m/s and the turbulence level is less than 0.1%. The model was placed in the middle of the test section and it spanned all the test section width.

The model is equipped with on external bearings to grant the possibility to continuously change the angle of attack by using a hydraulic actuator and it is connected to two external force balances as shown in Fig. 6 (left) in order to measure drag, lift and moment loads following the conventions reported in Fig. 7.

Drag (F_D), lift (F_L) and pitching moment (M) are provided in terms coefficients per unit deck length, C_D, C_L, C_M according to following formulation:

$$C_D(\alpha) = \frac{F_D}{q\ell B} \quad (2)$$

$$C_L(\alpha) = \frac{F_L}{q\ell B} \quad (3)$$

$$C_M(\alpha) = \frac{M}{qB^2\ell} \quad (4)$$

where B and ℓ represent the deck chord and the length of the model, respectively, and α is the angle of attack.

The model was also instrumented with laser transducers to monitor mid span displacement and mean angle of attack and with accelerometers to measure vertical and horizontal accelerations. This set-up permitted to measure also the static deformation of the model due to its flexibility.

Set-up 2 (see Fig. 5 (right)) was mounted in the boundary layer tests section of the wind tunnel: the test section is 4 m wide and 14 m high, the maximum wind velocity is 15 m/s and the turbulence level is less than 2%.

The sectional model was elastically suspended on stays and springs: the suspension system was designed to grant the lowest possible damping ratio, i.e. to have the lowest possible Scruton Number.

This experimental set-up was used to investigate the vortex-induced vibrations in vertical direction; the natural vertical frequency of the model ($f_N = 2.74$ Hz) was tuned to have the lock-in velocity at about 2 m/s ($V^* = 0.5$) and at about 3.5 m/s ($V^* = 0.8$). The reduced

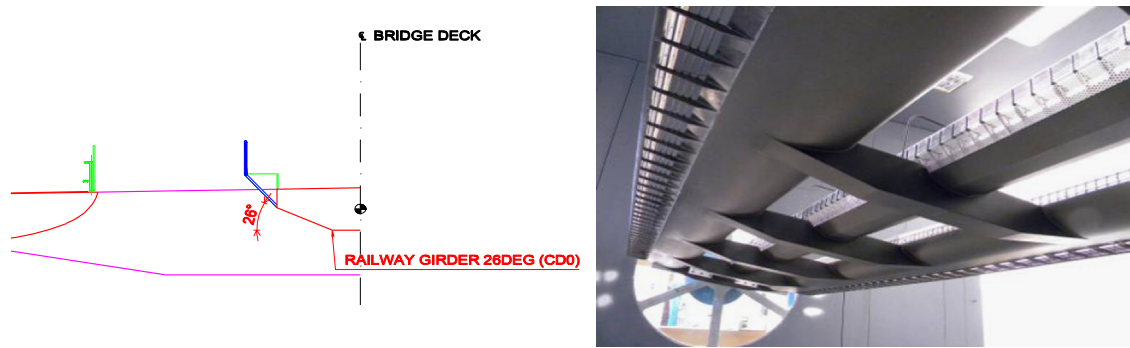


Fig. 2. MB01: railway girder lateral side slope 26 deg.

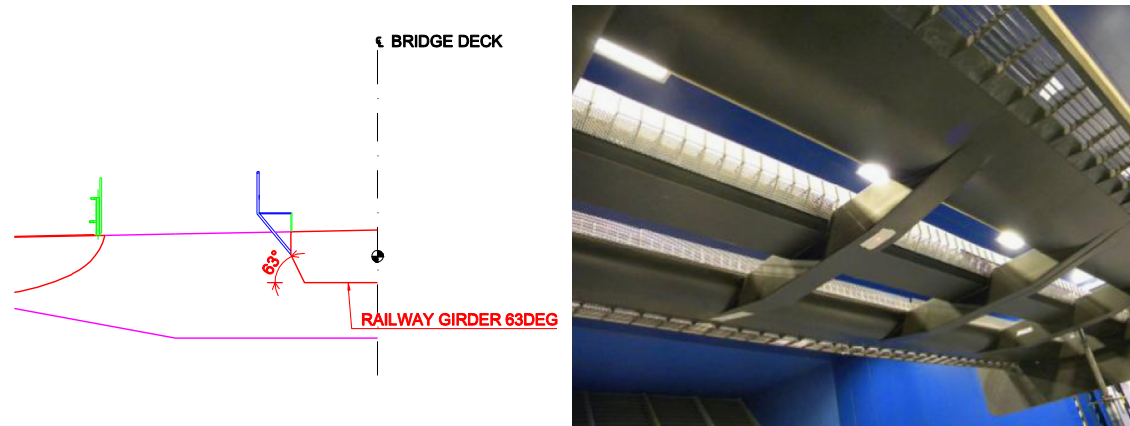


Fig. 3. MB021: railway girder lateral side slope 63 deg.

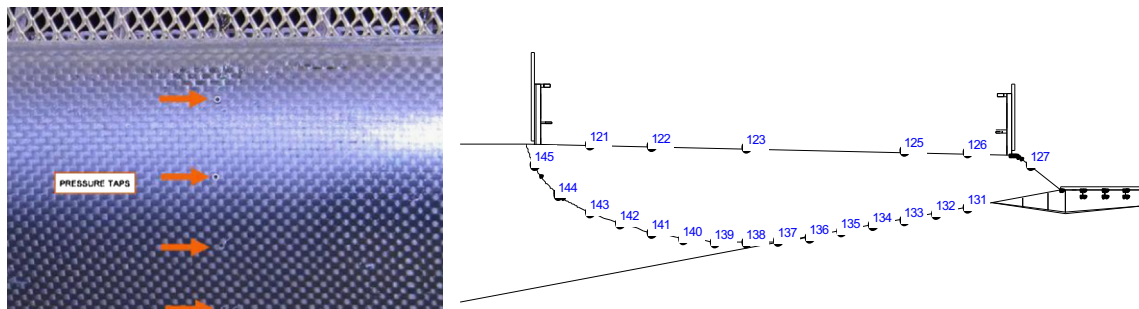


Fig. 4. (left) Detail of the lower side of the roadway girder, the pressure taps are highlighted (right) pressure taps distribution on the downwind roadway girder.

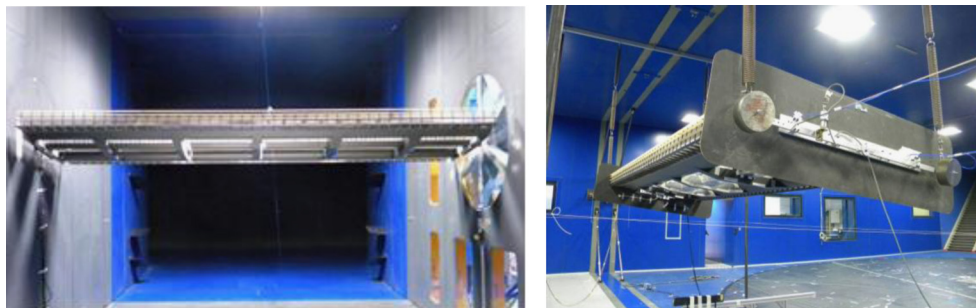


Fig. 5. Deck sectional model of the Messina Strait bridge in the Politecnico di Milano Wind tunnel: (left) dynamometric set-up (right) suspended set-up.

velocity is defined as $V^* = U/(f_N B)$ where B is the deck chord. In order to evaluate the phenomenon in different Scruton Number conditions, eddy current dampers were installed in different positions to increase the non dimensional damping ratio of the system. The dynamic behaviour of the suspended model was monitored through laser transducers and accelerometers.

3. Experimental results

Two models have been tested in both the over mentioned set-ups.

Set-up 1 permitted to measure, through the external balances, the aerodynamic forces on the deck as a function of the angle

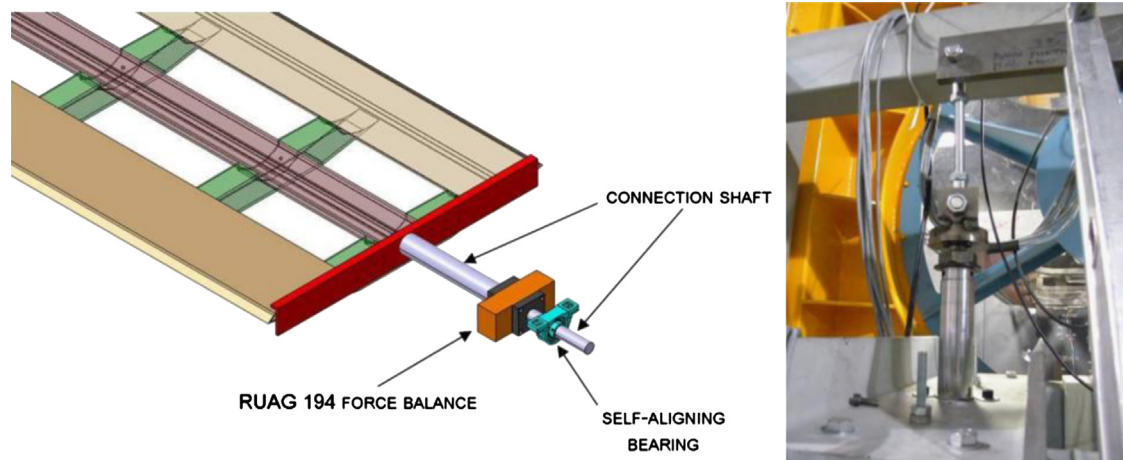


Fig. 6. (left) Detail of the external balance mounting (right) actuator for the model rotation.

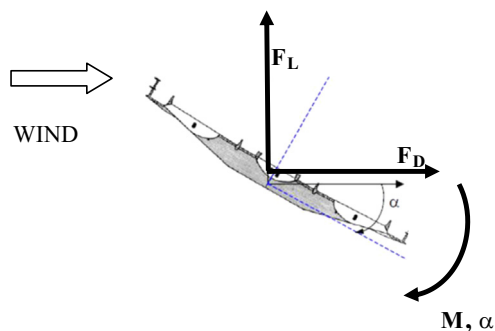


Fig. 7. Forces and angle of attack conventions.

of attack and for different wind velocities. Fig. 8 shows, as an example, the mean values of drag, lift and moment coefficients measured for a wind velocity equal to 12–15 m/s: no significant differences have been highlighted changing the wind velocity up to 30 m/s, resulting in a Reynolds number equal to 2.6×10^6 .

Lift and moment coefficients are characterized by positive derivatives for all the tested angle of attack for both models that show a very similar behaviour in terms of static aerodynamic forces: the positive slope is very important to avoid one-degree-of-freedom instability.

The overall forces seems to be slightly affected by the railway girder shape. Vortex shedding phenomenon was then investigated, in particular the harmonic content of the surface pressure distributions was considered.

It has been observed that, at 0 deg as angle of attack, the phenomenon is present for the model MB01 at about 5 m/s and it can introduce enough power in the system to excite the first flexural frequency of the model at $V^* = 0.8$ ($f_{flex} = 4.21$ Hz).

Fig. 9 shows the pressure distribution measured on the model MB01 when it experiences vortex shedding at $V = 4.64$ m/s. The dynamic part of the pressure coefficient fluctuations measured at the vortex shedding frequency is reported. The arrows lengths are proportional to the value of the peak pressure coefficient fluctuation. In this conditions the mid-span of the model was moving of about 1.33 mm i.e. a nondimensional displacement of 0.001 normalized on the chord B.

This representation is useful to appreciate how the vortex shedding mechanism is generated. The separation point moves along the upwind box and the vortex street is driven by the rail box (central one). Fig. 9 highlights that there is a strong pressure

fluctuation in the gap between the downwind roadway girder and the railway girder. The flow in the gap between the central and the downwind boxes strongly influences the vortex wake. The fluctuating flow between the two girders was observed also through smoke visualization: Fig. 10 shows four screenshots caught during the tests. This kind of vortex generation was observed also in Kwok et al. (2012) for a medium/high level of gap width between two twin decks. Kwok identified three sources of vortex shedding for twin-deck bridge: the vortices can shed from the trailing edge of the upstream deck, from the trailing edge of the downstream deck or they can be generated by the gap between the two decks. In this last situation the downstream deck is immersed in the wake of the upstream one and large negative pressures are observed on it due to separated flow. In the present case the gap between the downstream roadway girder and the railway girder seemed to be responsible of the observed vortex shedding and the high level of negative pressure measured on the downstream deck suggest that it can be considered immersed in separated flow.

Looking at the power spectrum of one selected pressure tap on the downwind roadway girder (taps 144 in agreement with the numeration reported in Fig. 4) a sharp peak at 4.21 Hz is clearly identified that corresponds to the flexural vibration frequency synchronized by the vortex shedding mechanism (see Fig. 11 (left)). A second power spectrum is reported with reference to an higher wind velocity (beyond the flexural mode lock-in region), where no vortex induced vibrations occurs (Fig. 11 (right)). In this situation the vortex shedding peak is still visible at a frequency of about 6 Hz but the spectrum is wider than in the previous situation, because at this wind velocity the vibration is not synchronized by the vortex shedding mechanism.

For the second configuration, MB02, small vortex induced vibrations occur. Looking at the spectrum of the pressure tap 144 at 4.64 m/s (Fig. 12 (left)) a narrow peak is still visible, but the modulus is strongly lower than the one measured on the MB01 configuration. For this reason the shape of the rail box girder has been evaluated fundamental in influencing vortex shedding phenomenon. At higher wind velocity (Fig. 12 (right)) a broad-band spectrum distributed around 6 Hz is visible. Model MB02 seems to experience vortex shedding as model MB01, but with a lower level of vibrations when synchronization occurs.

In order to better understand the flow–structure interaction for both the models, the second set-up was designed. Suspended set-up permit to evaluate the dynamic response on vertical direction of the sectional model, at different wind velocities. Only the 0 deg angle of attack was tested. The wind velocity was increased step by step and the dynamic behaviour of the deck was measured.

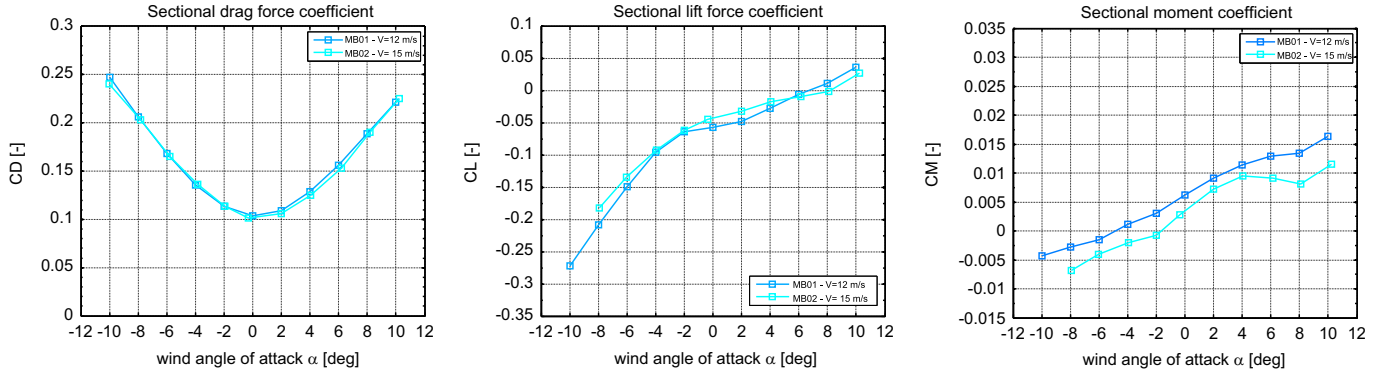


Fig. 8. Static force coefficients as a function of the wind angle of attack for both models.

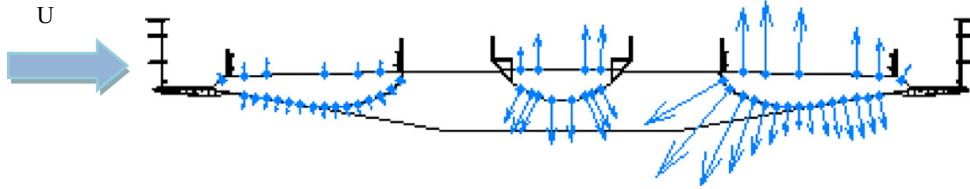


Fig. 9. Distributions of the peak spectra of the pressure coefficients (peaks have been evaluated in correspondence of the flexural frequency of the model).

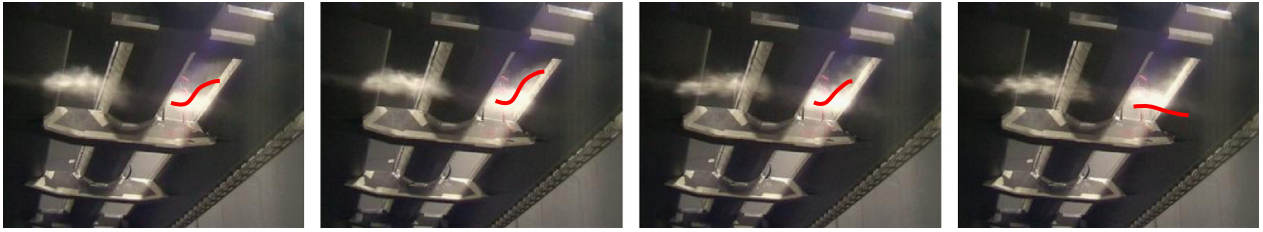


Fig. 10. Screen shots during the flow visualization tests.

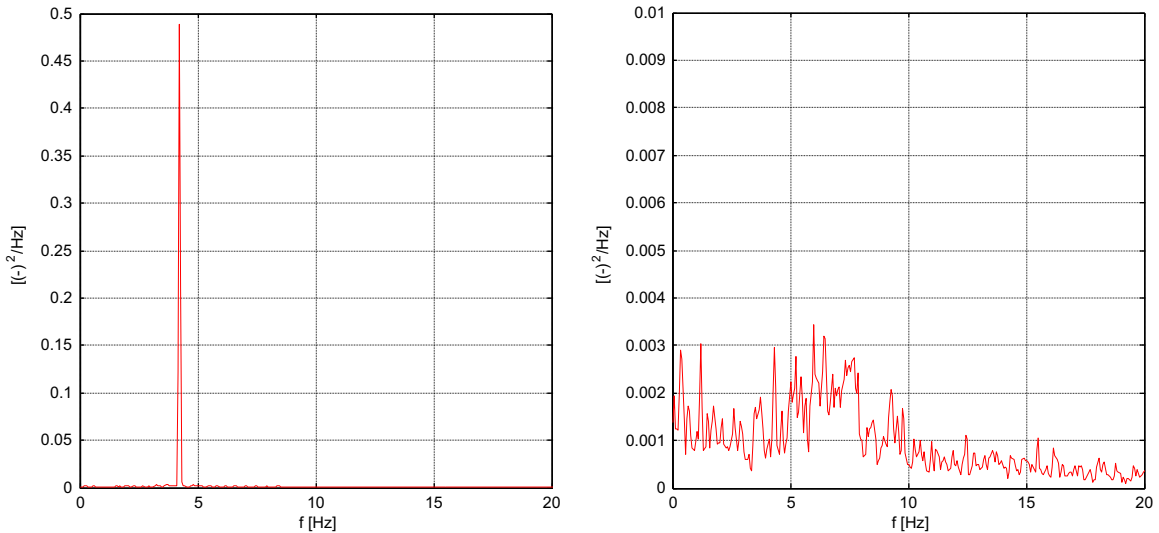


Fig. 11. Power spectrum of the pressure signal in correspondence of the tap 144 on model MB01 $V=4.64$ m/s (left) $V=6.79$ m/s (right).

Fig. 13 reports the steady state non-dimensional amplitudes (z/B) measured for different structural damping for the configurations MB01 (left) and MB02 (right). The responses have been evaluated as a function of the reduced wind velocity.

The different levels of damping considered are reported in Fig. 14 as a function of the oscillation amplitude of the model, z in terms of non-dimensional damping ratio (ξ_s). The reference Scruton numbers for each level are in Table 1, calculating the

number as

$$Sc = \frac{2\pi\xi_s m_l}{\rho B^2}$$

where m_l is the mass per unit length equal to 14.7 kg/m.

While single box deck shapes generally shows a single mechanism of vortex shedding, basically related to the vortexes detached in the girder wake, multi box decks may present different vortex shedding

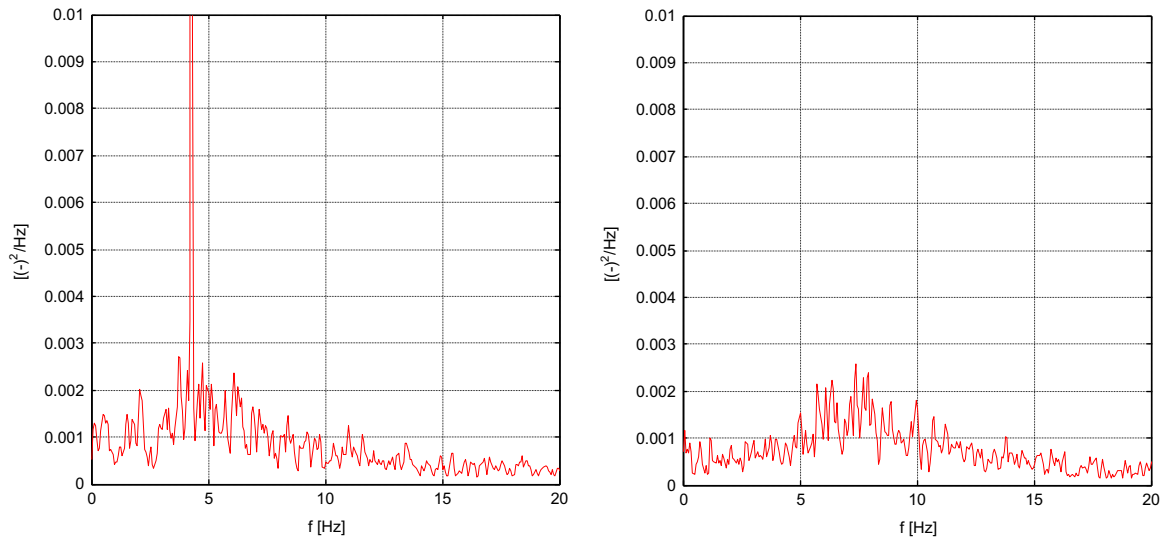


Fig. 12. Power spectrum of the pressure signal in correspondence of the tap 144 on model MB02 $V=4.64$ m/s (left) $V=6.79$ m/s.

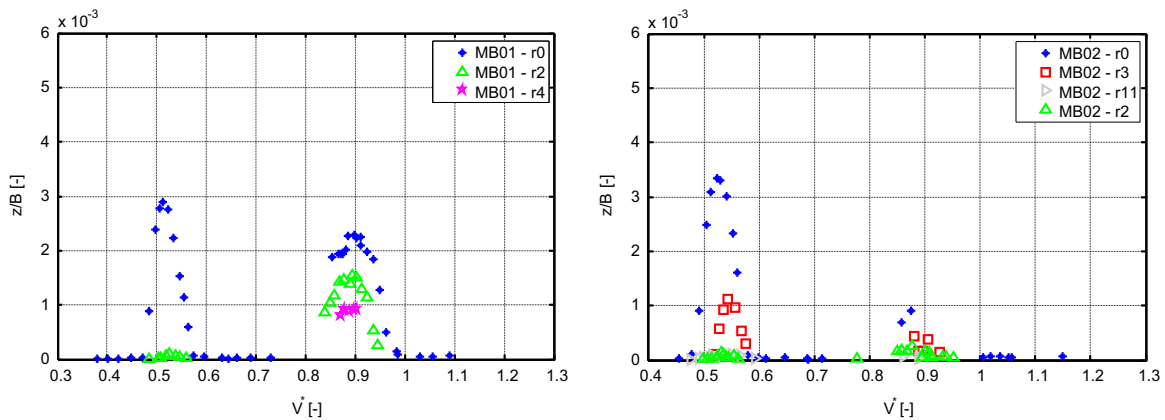


Fig. 13. Steady state response: non dimensional oscillation amplitude, flexural motion as function of the reduced velocity varying the Scruton number (left) MB01 (right) MB02.

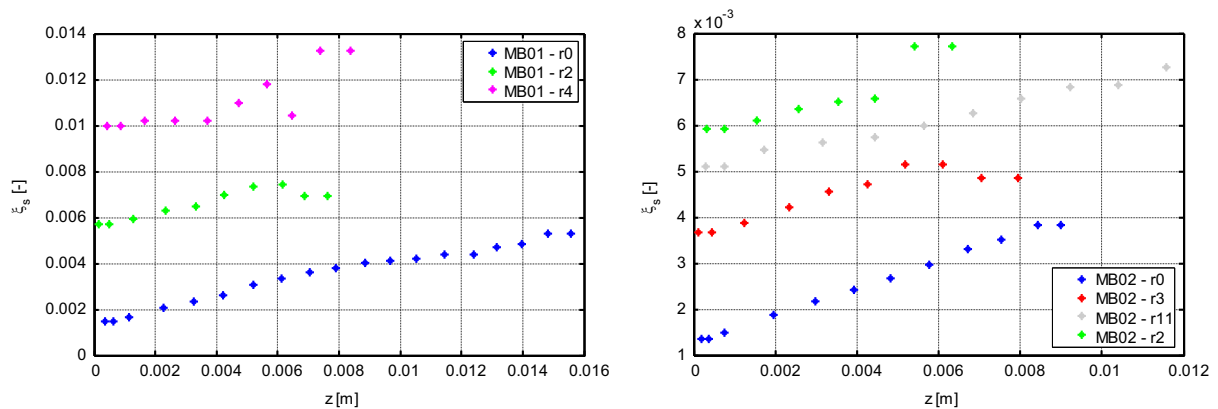


Fig. 14. Non-dimensional damping as a function of the oscillation amplitude z (left) MB01 (right) MB02.

phenomena because of the different possibility of flow interactions between the wake of the upwind box and the other boxes. The first lock in region ($V^* \approx 0.5$) was experienced by both models with similar levels of nondimensional vibration amplitude. However this region is very sensitive to damping increase and a non-dimensional damping equal to 0.5% is enough to reduce vortex induced vibrations to very low values (z/B equal to 10^{-4}). The second lock-in region ($V^* \approx 0.8$) is

more significant for model MB01 than for model MB02 and it is also less sensitive to the damping: for model MB01 a non-dimensional damping equal to 1% is not able to reduce the vibration amplitude under an acceptable threshold value.

On the other hand on model MB02, 0.5% of non-dimensional damping reduces also the vibrations related to the second lock-in region.

In Fig. 15 the steady state response of the model has been analysed considering, for some damping configurations, the lift force coefficient on the two roadway girders, upwind and downwind. These forces were obtained integrating the pressure taps along two sections of the girders and they are representative of the flow–structure interaction without taking into account the forces developed by the fittings. The reported data represent the amplitude of the oscillating Lift coefficient as a function of constant wind velocity: this oscillation is synchronized with the deck displacement.

As observed in terms of non-dimensional displacement, two lock-in regions can be identified. The first one is similar for the two models and can be easily controlled increasing non dimensional damping. The second one is more important for model MB01 and it is characterized by high level of lift coefficient, for the downwind girder. When vortex shedding occurs also the level of the fluctuating lift force synchronous with the displacement increase (see Zasso et al., 2008).

Pressure distributions have been considered in order to better highlight the two vortex shedding mechanism related to the identified lock-in regions. Figs. 16 and 17 show the peak spectra

distributions of the pressure coefficients for model MB01 and MB02, for two different reduced wind velocities. Fig. 16(b) concerns the second lock-in region and is quite similar to Fig. 9 with high pressures on the downwind roadway girder. The same vortex shedding mechanism previously described is observed. Fig. 16 (a) concerns the first lock-in region and shows higher pressures in the upwind roadway girder than in the downwind one. The source of the vortex shedding in this case can be considered the upwind roadway girder where high negative pressures were measured, whereas the wake effects on the downwind girder are less important (see Kwok et al., 2012).

Model MB02 in correspondence of the first lock in region shows a pressure distribution similar to the one observed for model MB01 (Fig. 17 (a)): in both cases, high suction were measured especially in the upwind road girder. This confirms the results observed in terms of dynamic response of the two models in correspondence of the first lock in region. On the other hand, in correspondence of the second lock in region, model MB02 experienced very low values of surface pressures: in particular the downwind road girder is characterized by suction three times lower with respect to the ones measured on model MB01. This behaviour confirms the different attitudes of the models to the second vortex shedding mechanism highlighting the more advisable behaviour of model MB02 compared to model MB01, in terms of dynamic response.

Table 1

Scruton numbers for the different tested configurations, calculated in correspondence of small oscillation amplitude.

MB01		MB02	
Configuration	Scruton number	Configuration	Scruton number
r0	0.07	r0	0.06
r2	0.26	r3	0.16
r4	0.44	r11	0.23
		r2	0.26

4. Conclusions

This paper describes selected results collected during the wind tunnel tests carried out at Politecnico di Milano Wind Tunnel on the deck sectional model of the Messina Strait Bridge.

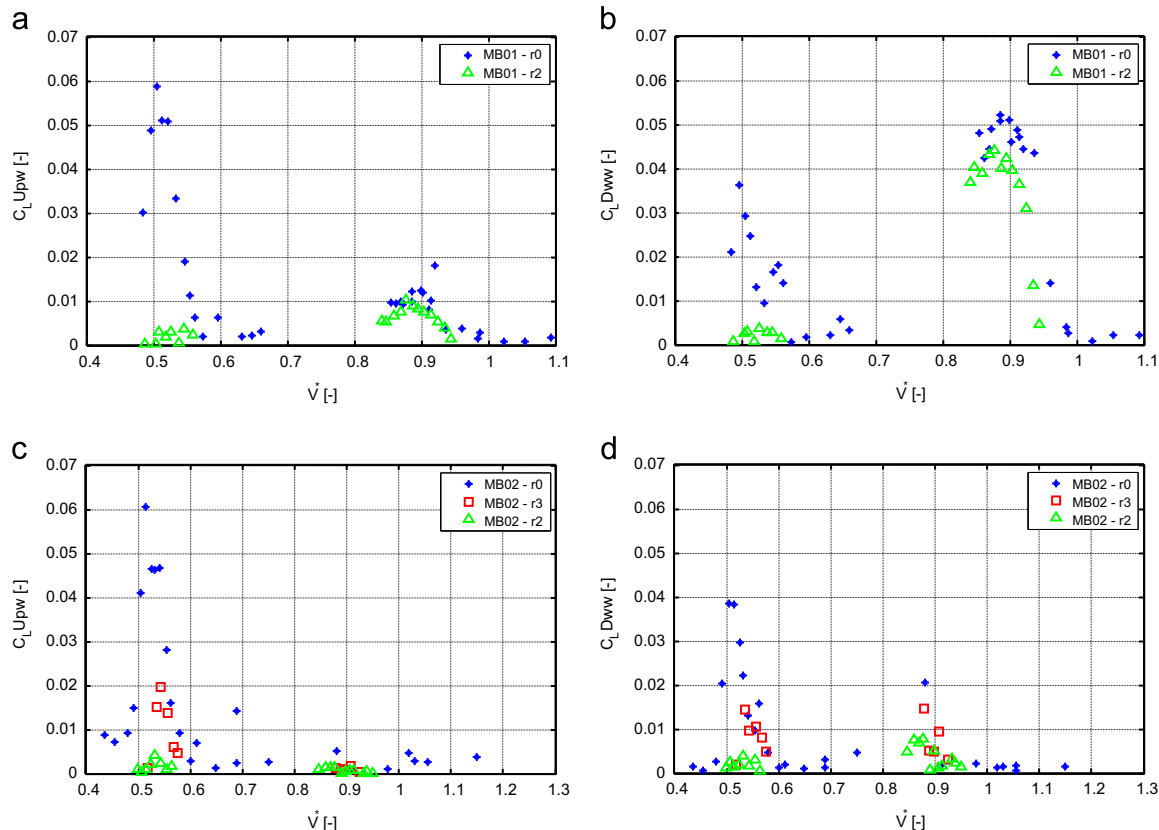


Fig. 15. Steady state lift coefficient on the road girders as function of the reduced velocity varying the Scruton number (a) MB01, lift coefficient on the upwind road girder (b) MB01, lift coefficient on the downwind road girder (c) MB02, lift coefficient on the upwind road girder (d) MB02, lift coefficient on the downwind road girder.

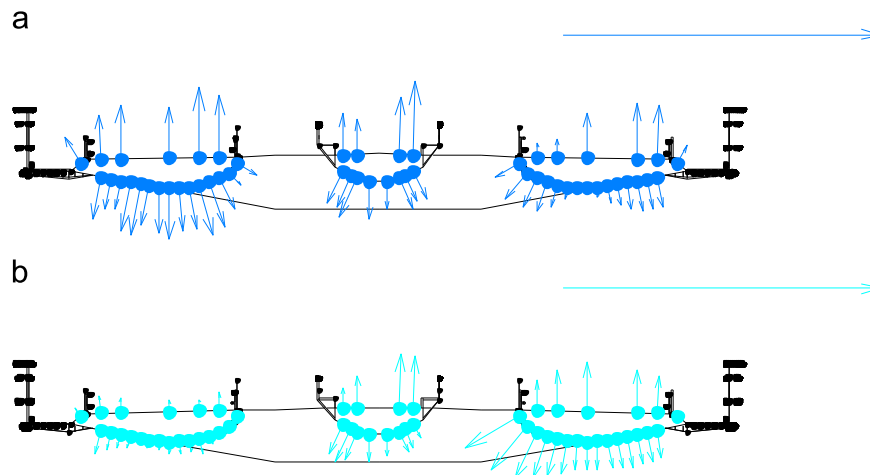


Fig. 16. Distributions of the peak spectra of the pressure coefficients on model MB01 (peaks have been evaluated in correspondence of the model natural frequency) (a) $V^* = 0.5$ –(b) $V^* = 0.87$.

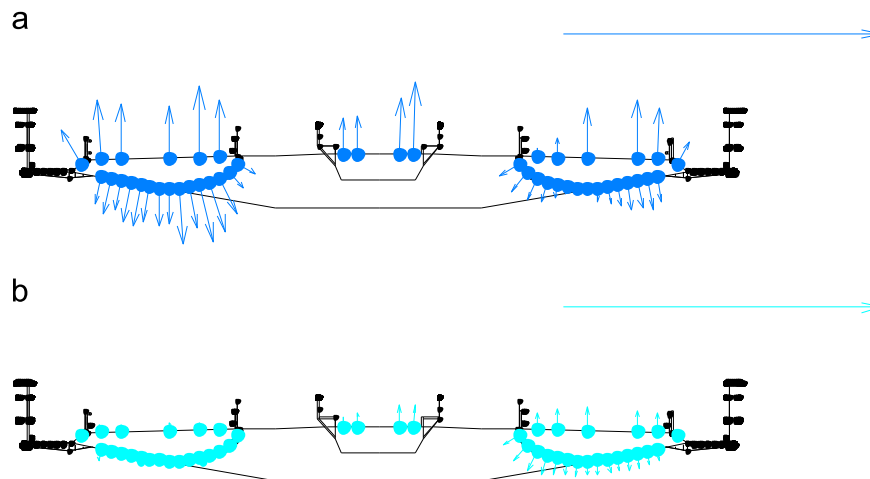


Fig. 17. Distributions of the peak spectra of the pressure coefficients on model MB02 (peaks have been evaluated in correspondence of the model natural frequency) (a) $V^* = 0.53$ –(b) $V^* = 0.87$.

The tests were carried out both in the high speed than in boundary layer test section of the wind tunnel. Two set-ups were built, one with external balances and an elastically suspended one. The suspension system was designed to grant the lowest possible damping ratio, i.e. to have the lowest possible Scruton Number. Moreover, it was decided to tune the natural frequencies in order to have the lock-in velocity over 2 m/s, to grant the needed flow stability. All these technical solutions were chosen in order to maximize vortex-induced vibrations. Two rigid sectional model were tested to check the effect of the central railway girder shape on vortex induce vibration phenomenon. The measurement on the MB01 original solution highlighted very high vortex shedding vibrations and the pressure measurements demonstrate that different kind of vortex shedding mechanism can be experienced by a multi-box deck. In the most critical condition the vortex induced vibrations are related to the flow in the gap between the rail and the road boxes: changing the angle of the rail box girder changes the air intake into the gap modifying the vortex wake and decreasing the pressure fluctuations and the VIV in a large amount.

To define the vibration amplitude induced by vortex shedding on the real bridge, the non-dimensional amplitudes have been reported as a function of the Scruton Number. As can be seen for a value of structural damping equal to 0.5% the non-dimensional vibration amplitudes (z/B) are lower than the value $1 \times 10^3 z/B$ or 6 mm. On the other hand, considering a value of structural

damping of 0.3% the non-dimensional amplitudes (z/B) are equal to 4×10^3 , or 24 mm at full scale.

References

- Argentini, T., Rocchi, D., Muggiasca, S., Zasso, A., 2012. Cross-sectional distributions versus integrated coefficients of flutter derivatives and aerodynamic admittances identified with surface pressure measurement. *J. Wind Eng. Ind. Aerodyn.* 104–106, 152–158.
- Belloli, M., Fossati, F., Giappino, S., Muggiasca, S., Villani, M., 2011. On the aerodynamic and aeroelastic response of a bridge tower. *J. Wind Eng. Ind. Aerodyn.* 99, 729–733.
- Belloli, M., Giappino, S., Muggiasca, S., Zasso, A., 2012. Force and wake analysis on a single circular cylinder subjected to vortex induced vibrations at high mass ratio and high Reynolds number. *J. Wind Eng. Ind. Aerodyn.* 103, 96–106.
- Caracoglia, L., Noe, S., Sepe, V., 2008. Nonlinear Computer Model for the Simulation of Lock-in Vibration on Long-Span Bridges. *Computer-Aided Civil Infrastruct. Eng.* 24, 1–15.
- Diana, G., Zasso, A., Resta, F., Belloli, M., Rocchi, D., 2004. Forced motion and free motion aeroelastic tests on a new concept dynamometric section model of the Messina suspension bridge. *J. Wind Eng. Ind. Aerodyn.* 92, 441–462.
- Diana, G., Resta, F., Belloli, M., Rocchi, D., 2006. On the vortex shedding forcing on suspension bridge deck. *J. Wind Eng. Ind. Aerodyn.* 94, 341–363.
- Fossati, F., Muggiasca, S., Giappino, S., Squicciarini, G., Ozkan, E., Belloli, M., 2010. Wind induced vibrations on Bridge Towers: vortex shedding investigation using sectional models. In: *Proceedings of the Ninth UK Conference on Wind Engineering WES*, Bristol UK, September 20–22 2010.
- Kwok, K.C.S., Qin, X.R., Fok, C.H., Hitchcock, P.A., 2012. Wind-induced pressures around a sectional twin-deck bridge emodel: effects of gap-width on the

- aerodynamic forces and vortex shedding mechanisms. *J. Wind Eng. Ind. Aerodyn.* 110, 50–61.
- Laima, S., Li, H., Chen, W., Li, F., 2013. Investigation and control of vortex-induced vibration of twin box girders. *J. Fluids Struct.* 39, 205–221.
- Larsen, A., Esdahl, S., Andersen, J.E., Vejrum, T., 2000. Storebælt suspension bridge—vortex shedding excitation and mitigation by guide vanes. *J. Wind Eng. Ind. Aerodyn.* 88, 283–296.
- Larsen, A., Wall, A., 2012. Shaping of bridge box girders to avoid vortex shedding response. *J. Wind Eng. Ind. Aerodyn.* 104–106, 159–165.
- Li, H., Laima, S., Oua, J., Zhaob, X., Zhoua, W., Yub, Y., Li, N., Liu, Z., 2011. Investigation of vortex-induced vibration of a suspension bridge with two separated steel box girders based on field measurements. *Eng. Struct.* 33, 1894–1907.
- Zasso, A., Belloli, M., Giappino, S., Muggiasca, S., 2008. Pressure field analysis on oscillating circular cylinder. *J. Fluids Struct.* 24, 628–650.